

Does National Individualism Predict AI Sentiment?

Testing the Cultural-Dimensions Explanation for the Asia-West AI Optimism Gap

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Abstract

Global surveys of public attitudes toward AI consistently find a striking regional pattern: countries such as Indonesia and Thailand report overwhelming optimism about AI, while countries such as Canada and the United States are far more skeptical. Cross-cultural psychology offers a candidate explanation: individualistic cultures tend to frame new technology as a potential threat to personal autonomy, while collectivist cultures more often frame it as a tool that helps people function within their social environment. This analysis tests that explanation directly and quantitatively, correlating Hofstede's Individualism Index against Ipsos's 2025 AI Monitor survey (30 countries, $n=23,216$ adults). The result is a strong, statistically robust negative correlation ($r=-0.85$, $p=2.8 \times 10^{-9}$): individualism alone explains approximately 72% of the cross-country variation in AI sentiment. The relationship

survives a robustness check excluding six countries whose cultural-dimension data could not be confirmed against a single consistent source.

1 Introduction and Motivation

Public commentary on the global divide in AI sentiment tends to reach for economic explanations (wealthier countries have more to lose; poorer countries see AI as a path to catching up) or narrower cultural anecdotes (Japan’s friendly-robot pop culture versus Hollywood’s dystopian AI narratives). This analysis tests a more specific, established psychological construct directly: Hofstede’s Individualism Index, a widely used measure of whether a society emphasizes individual autonomy or group interdependence. The hypothesis, drawn from published cross-cultural psychology research, is that individualistic societies frame AI primarily as a threat to personal autonomy and privacy, while collectivist societies more often frame it as an extension of the self that helps one function within a group – a positive frame in a collectivist value system.

2 Research Question

Does Hofstede’s Individualism Index (IDV) correlate with the percentage of a country’s population who see AI as more beneficial than harmful?

3 Data and Limitations

AI sentiment. Ipsos AI Monitor 2025, a 30-country Global Advisor survey of 23,216 adults, fielded March 21 - April 4, 2025. The specific item used is “Products and services using artificial intelligence have more benefits than drawbacks.”

Cultural dimension. Hofstede’s Individualism Index (IDV), a 0-100 scale where higher values indicate greater individualism.

Known limitations.

- **This Ipsos wave does not include China.** The widely cited “China 83%” AI-optimism figure comes from a different Ipsos wave (cited in Stanford’s AI Index 2025), not the 2025 30-country wave used here. Ipsos’s country coverage changes between waves, so a single-source, multi-year, China-inclusive panel is not available from this survey series alone.
- **Hofstede IDV source consistency.** Twenty-four of the 30 countries’ IDV scores come from a single, internally consistent peer-reviewed table. The remaining six (Thailand, Singapore, Ireland, Sweden, Hungary, Spain) are drawn from supplementary sources and could not be confirmed against that same table. This matters because Hofstede’s published scores have been revised over time – the official source notes that the individualism and long-term-orientation dimensions were last updated in October 2023 – so mixing values from different vintages across countries is a genuine, disclosed risk. This is addressed directly with a robustness check (Section 5) that re-runs the analysis excluding these six countries.

- **Sample representativeness.** Ipsos’s own methodology discloses that several countries in the survey (including Brazil, Indonesia, Malaysia, Thailand, and Turkiye) have samples skewed toward more urban, educated, or affluent respondents than their general population. Results for these countries should be read as reflecting a more “connected” segment of the population.
- **Correlational, not causal.** A strong correlation between individualism and AI sentiment does not establish that cultural values cause the sentiment gap. Both could be driven by a shared third factor – for example, economic growth expectations, government AI strategy and messaging, or media environment – that this analysis does not measure directly.

4 Methodology

1. **Clean and merge** (`analysis.R`, Part 1): the two datasets were joined on country name. All 30 Ipsos countries had an available Hofstede IDV score. Each row was flagged according to whether its IDV score came from the single consistent source or a supplementary source.
2. **Test** (Part 2): Pearson and Spearman correlations were computed between IDV and AI sentiment, followed by a simple linear regression. A robustness check reran the correlation using only the 24 countries with verified-consistent IDV data, to confirm the result was not being driven by the six less-certain data points.

5 Results

Following the merge, the analysis dataset contains all 30 Ipsos-surveyed countries with a matched Hofstede IDV score.

Table 1: Summary of statistical tests.

Test	Result
Pearson correlation	$r = -0.85$, $p = 2.8 \times 10^{-9}$
Spearman correlation	$\rho = -0.85$, $p = 1.8 \times 10^{-9}$
Linear model	$R^2 = 0.72$, slope = -0.42
Robustness check (n=24, verified-source only)	$r = -0.83$, $p = 5.2 \times 10^{-7}$

The relationship is visualized in Figure 1: a strong, clearly linear, negative relationship, with the most collectivist countries in the sample (Indonesia, Thailand, Singapore) clustered at the high-sentiment end, and the most individualistic countries (the United States, Australia, the United Kingdom, Canada) clustered at the low-sentiment end.

Individualism alone explains approximately 72% of the variance in AI sentiment across these 30 countries ($R^2=0.72$) – a substantially larger effect than is typical in cross-national social-attitude research, where a single predictor explaining most of the variance in a complex attitude is unusual. The robustness check confirms this

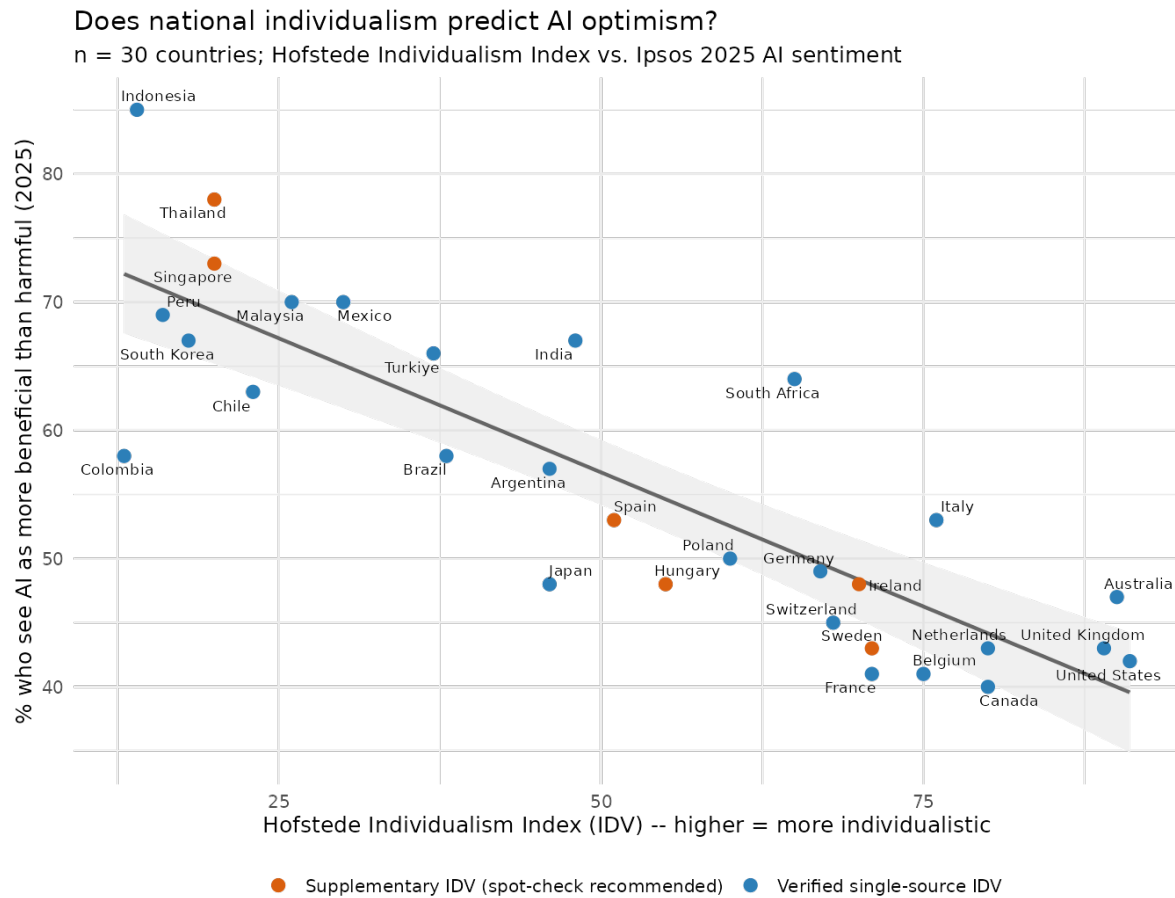


Figure 1: Hofstede Individualism Index vs. Ipsos 2025 AI sentiment, 30 countries. Blue points use IDV scores confirmed against a single consistent source; orange points use supplementary sources.

is not an artifact of the six less-certain data points: restricting the sample to the 24 countries with fully verified IDV scores still yields $r=-0.83$, $p=5.2 \times 10^{-7}$ – a negligible change from the full-sample result.

6 Discussion

The strength of this correlation ($R^2=0.72$) is worth treating with appropriate caution precisely because it is so strong. Cross-national social-attitude research rarely finds a single cultural variable explaining nearly three-quarters of the variance in a complex, multiply-determined attitude like technology sentiment. Three considerations are worth holding alongside the headline number.

First, individualism is very likely correlated with other national characteristics that plausibly also shape AI sentiment – GDP per capita, media environment, government AI strategy and public messaging, and existing familiarity with AI tools through daily life. This analysis tests a bivariate relationship; it does not control for these other factors, and a genuinely rigorous causal claim would require doing so. The strength of the correlation found here is consistent with individualism being a real contributing factor, a proxy for several correlated factors, or some combination of both – this analysis alone cannot distinguish between these possibilities.

Second, the exclusion of China from this particular Ipsos wave is a meaningful gap. China is frequently cited as the country with the single highest AI optimism (83% in a separate Ipsos wave) and one of the most collectivist cultures in Hofstede’s original framework (IDV=20 in the source table used here). Its absence from this specific 30-country sample does not undermine the finding, since the pattern holds clearly across the 30 countries that are included, but a fully verified China data point would have strengthened the sample’s coverage at exactly the extreme end of both variables.

Third, as discussed in the Data and Limitations section, six of the thirty IDV scores carry a disclosed data-quality caveat. The robustness check addressing this (Section 5) found a negligible change when these six countries were excluded, which increases confidence that the result is not being driven by potentially inconsistent data – but it does not eliminate the underlying uncertainty about those six countries’ exact scores.

7 Limitations and Future Work

1. **Multivariate modeling.** A regression including GDP per capita, government AI strategy indicators, and media environment measures alongside individualism would test whether the cultural-dimension effect survives controlling for these plausible confounds, rather than relying on a bivariate correlation alone.
2. **A fully single-sourced Hofstede dataset.** Obtaining all 30 countries’ IDV scores directly from a single official source (rather than the current mix of one consistent table plus six supplementary lookups) would remove the remaining data-quality caveat entirely.
3. **Including China.** Locating a wave of the Ipsos AI Monitor (or a comparable survey) that includes both China and the other 30 countries in a single consis-

tent administration would allow the extreme end of both variables to be tested directly within one dataset.

4. **Longitudinal tracking.** Since Ipsos fields this survey annually, tracking whether the individualism-sentiment relationship strengthens, weakens, or holds steady as AI becomes more familiar globally would be a natural extension.

8 Conclusion

Hofstede's Individualism Index is a strong, statistically robust predictor of national AI sentiment across the 30 countries surveyed by Ipsos in 2025, explaining approximately 72% of the cross-country variance and surviving a robustness check against a known data-quality limitation. This is consistent with – though not sufficient on its own to prove – the cross-cultural psychology explanation that individualistic societies frame AI primarily as a threat to personal autonomy, while collectivist societies more often frame it as a tool that helps people function within their social environment. A genuinely causal claim would require controlling for the economic and media-environment factors that plausibly correlate with both individualism and AI sentiment.

References

- Ipsos. (2025). *The Ipsos AI Monitor 2025: A 30-country Ipsos Global Advisor Survey*. Retrieved from ipsos.com
- Hofstede, G. (1997). *Cultures and Organizations: Software of the Mind*. McGraw-Hill.
- Kelley, K. J., et al. (2021). Cross-cultural comparison of AI attitudes across 8 countries. Referenced via secondary academic sources on individualism and AI sentiment.